Supporting Information for

Hydrogen-assisted Mg intercalation into 2H-TaS_2

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1. Change in lattice parameters

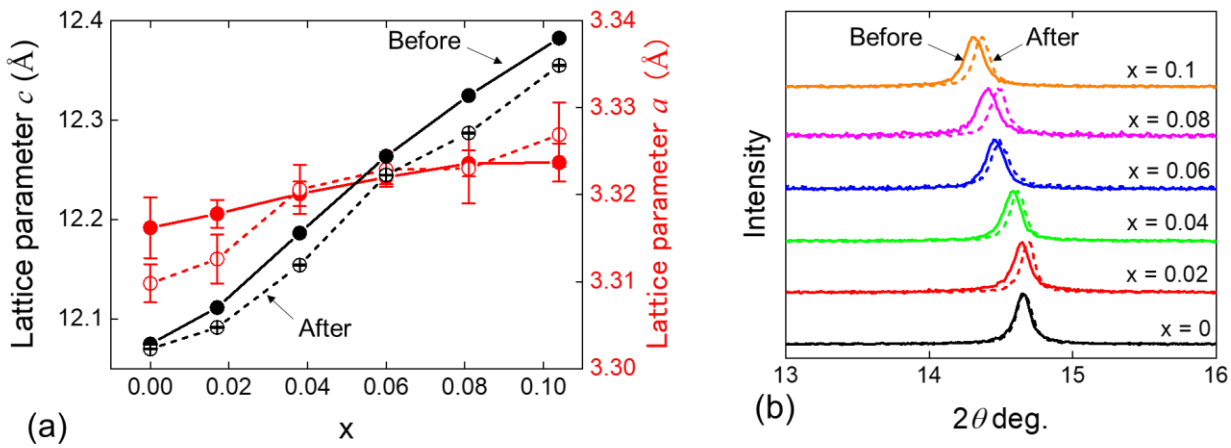


Figure S1. (a) Changes in lattice parameters with different Mg concentrations. The lattice parameters *c* and *a* are shown in black and red, respectively. Solid and dotted lines indicate the samples before and after H removal, respectively. (b) 002 XRD peaks for the samples with nominal x = 0, 0.02, 0.04, 0.06, 0.08, and 0.1. Solid and dotted lines indicate the samples before and after H removal, respectively.

Table S1. Lattice parameters with standard deviation for each sample

Mg	H	*a* (Å)	σ(*a*)	*c* (Å)	σ(*c*)
0.000	0.006	3.3162	0.0035	12.0746	0.0006
0.000	0.000	3.3098	0.0022	12.0698	0.0004
0.017	0.013	3.3178	0.0016	12.1114	0.0003
0.017	0.000	3.3126	0.0028	12.0914	0.0005
0.038	0.027	3.3201	0.0014	12.1864	0.0002
0.038	0.000	3.3206	0.0028	12.1541	0.0005
0.060	0.050	3.3220	0.0008	12.2637	0.0001
0.060	0.000	3.3229	0.0020	12.2449	0.0004
0.081	0.061	3.3236	0.0015	12.3248	0.0003
0.081	0.000	3.3230	0.0040	12.2871	0.0007
0.104	0.094	3.3237	0.0022	12.3826	0.0004
0.104	0.000	3.3269	0.0037	12.3551	0.0007

2. TPD measurements

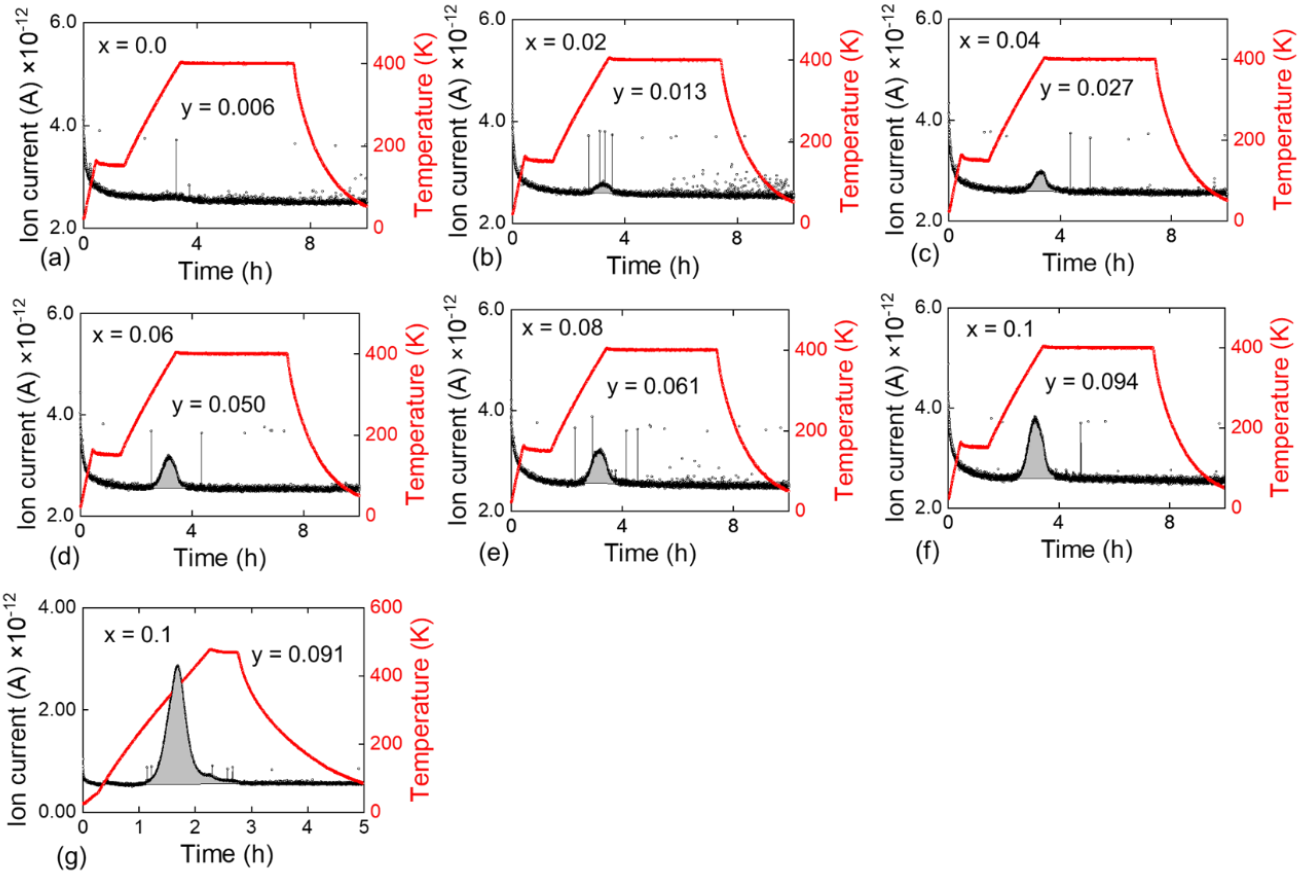


Figure S2. (a)–(f) TPD measurements in a N_2 gas flow for $Mg_xH_yTaS_2$ prepared at 650 °C under 1 GPa. (g) TPD measurement in a N_2 gas flow for $Mg_xH_yTaS_2$ prepared at 800 °C under 1 GPa. x indicates the nominal Mg concentration. Black points indicate the intensity of detected H_2, and the red line indicates the temperature sequence. The H amount y can be estimated from the gray highlighted area.

3. Supercells for DFT calculations

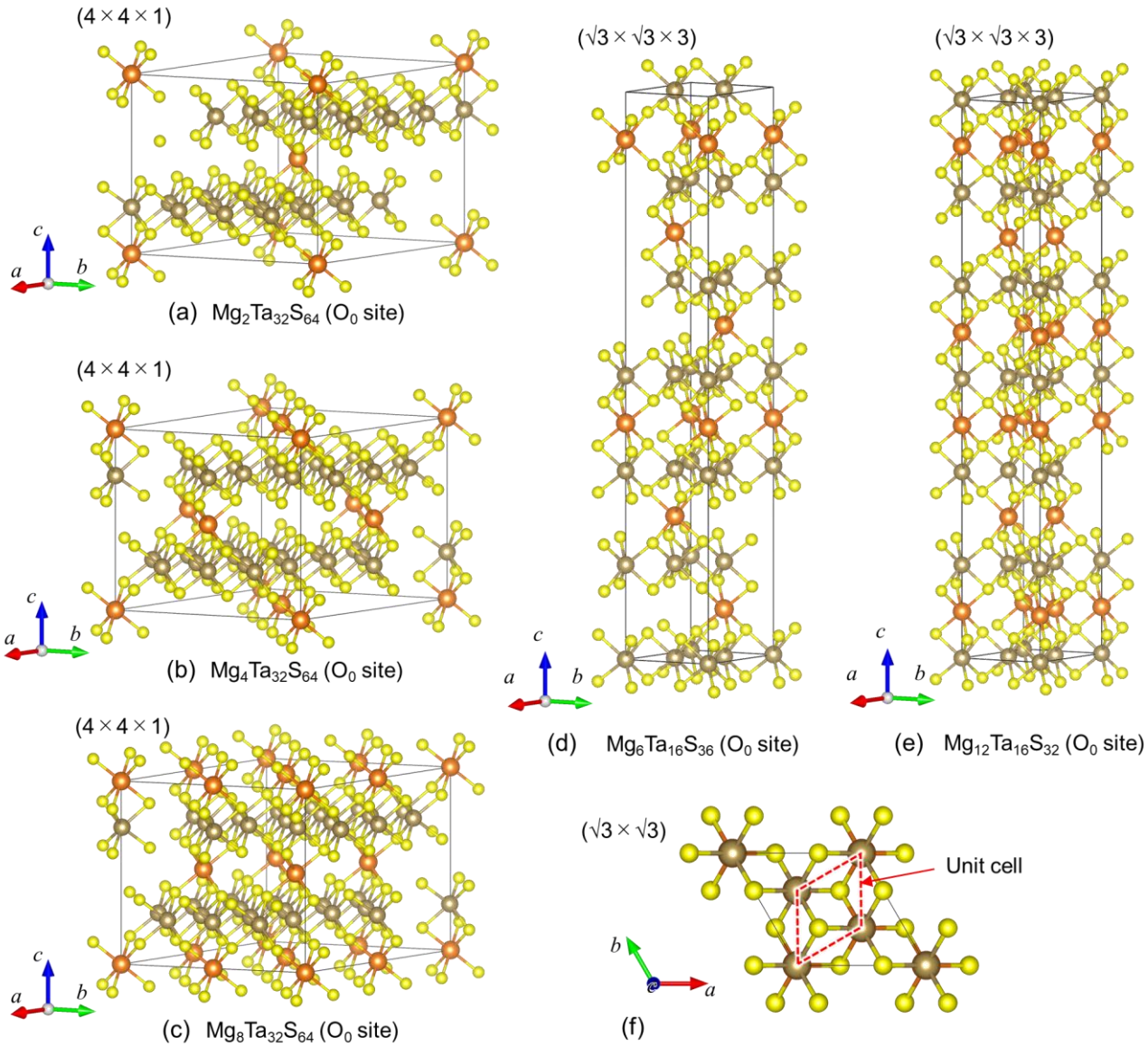
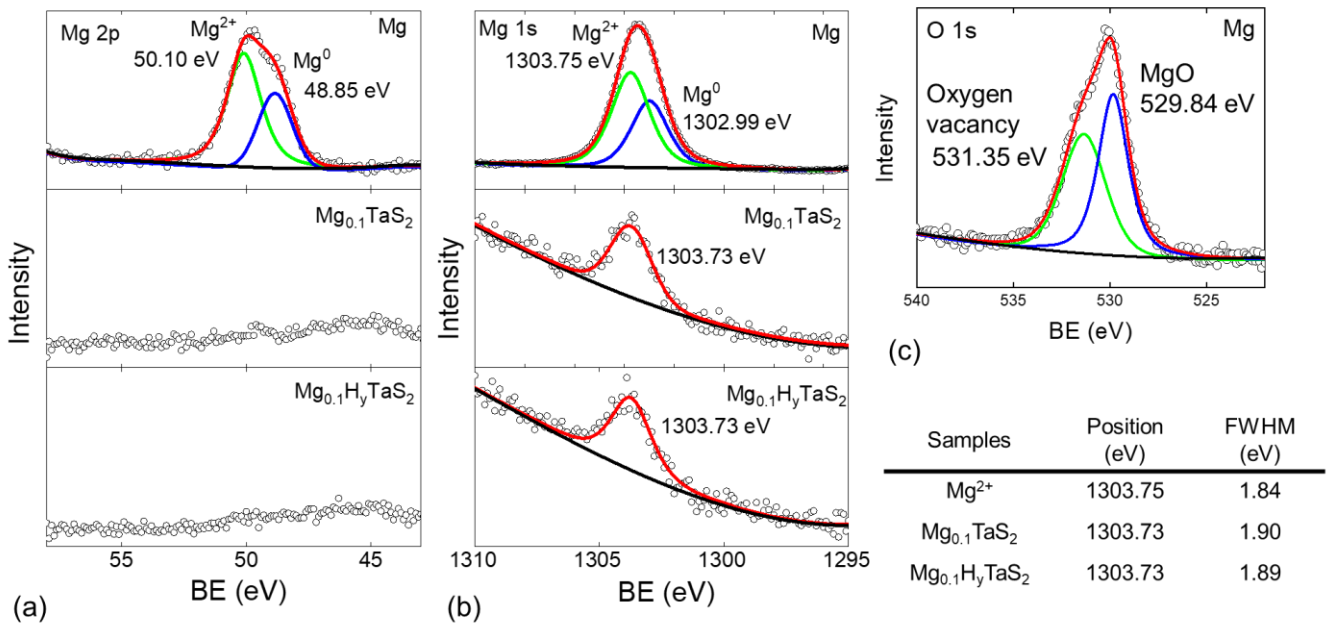


Figure S3. Crystal structures of supercells (4×4×1) with different Mg concentrations: (a) $Mg_2Ta_{32}S_{64}$, (b) $Mg_4Ta_{32}S_{64}$, and (c) $Mg_8Ta_{32}S_{64}$. Crystal structures of supercells (√3×√3×3) with different Mg concentrations: (d) $Mg_6Ta_{16}S_{32}$, and (e) $Mg_{12}Ta_{16}S_{32}$. (f) Supercells (√3×√3) from the *c*-axis direction.

4. The valence of guest ions



Samples	Position (eV)	FWHM (eV)
Mg^{2+}	1303.75	1.84
$Mg_{0.1}TaS_2$	1303.73	1.90
$Mg_{0.1}H_yTaS_2$	1303.73	1.89

Figure S4. (a) Mg 2p, and (b) Mg 1s XPS profiles for Mg, Mg_xTaS_2, and $Mg_xH_yTaS_2$ (x = 0.1). The black, red, green, and blue lines denote the baseline, total fitting curve, and the fitting curves assigned to Mg^{2+} and Mg^0, respectively. (c) O 1s XPS profile for Mg. The black, red, green, and blue lines denote the baseline, total fitting curve, and fitting curves assigned to MgO and oxygen vacancy, respectively.

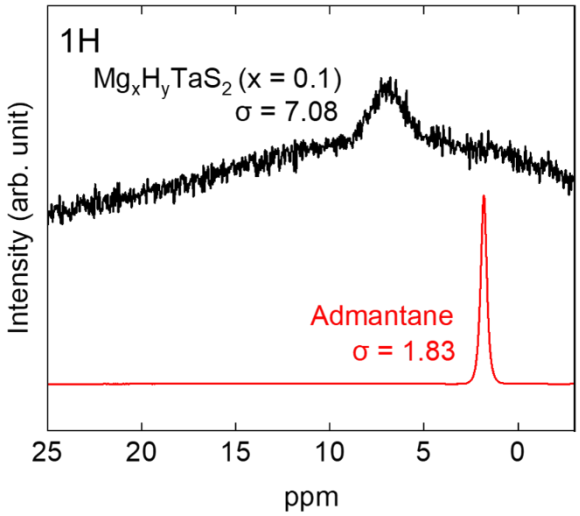


Figure S5. 1H NMR Spectra of $Mg_xH_yTaS_2$ (x = 0.1) and external standard of Admantane. Magic angle spinning rates for $Mg_xH_yTaS_2$ and Admantane are 45 kHz and 30 kHz, respectively.

5. Resistivity measurements and XRD

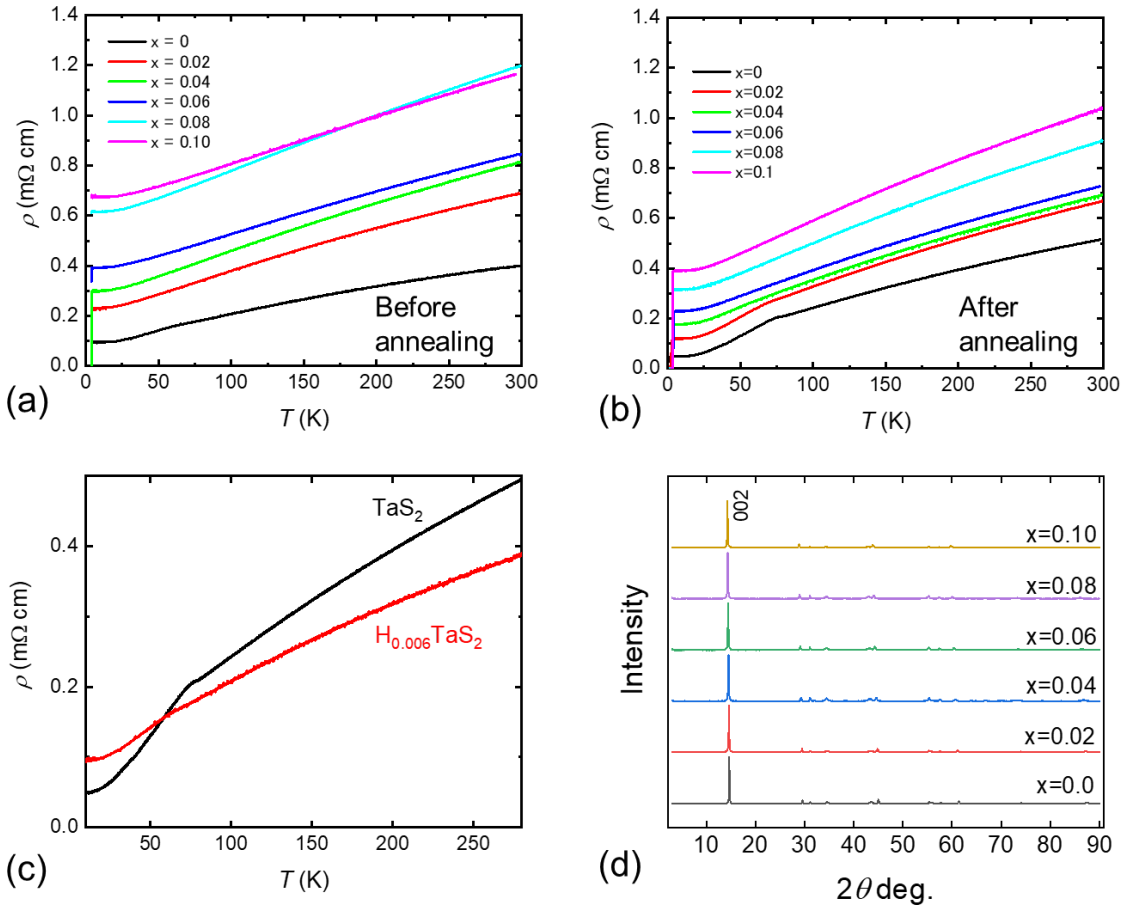


Figure S6. Temperature dependence of the resistivity for samples with different nominal Mg concentrations (x) (a) before and (b) after H removal. (c) Temperature dependence of the resistivity for TaS_2 and $H_{0.006}TaS_2$. (d) XRD profiles for $Mg_xH_yTaS_2$ with different x values.

6. Resistivity under magnetic field

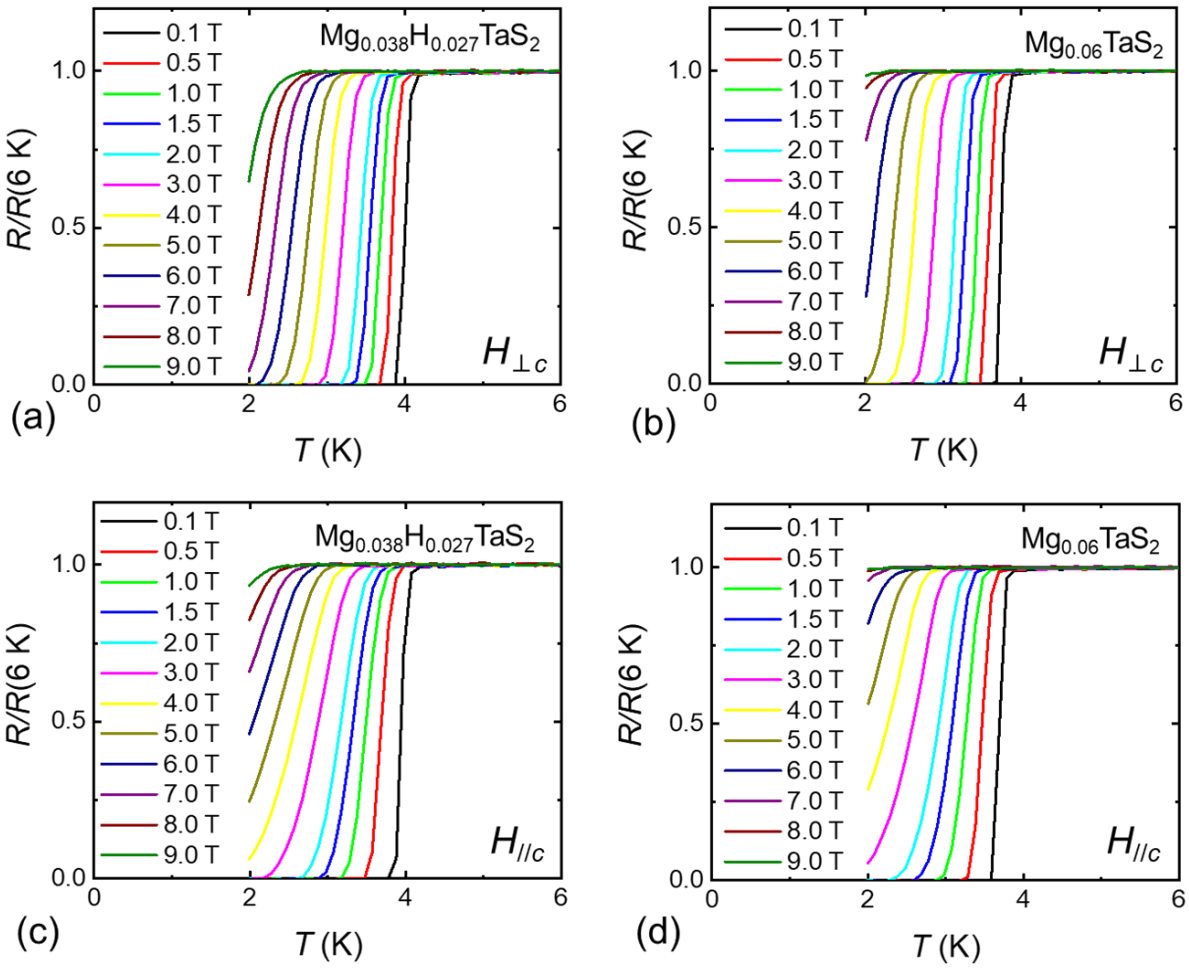


Figure S7. Temperature dependence of the normalized resistance for (a) $Mg_{0.038}H_{0.027}TaS_2$ under $H_{\perp c}$, (b) $Mg_{0.06}TaS_2$ under $H_{\perp c}$, (c) $Mg_{0.038}H_{0.027}TaS_2$ under $H_{//c}$, and (d) $Mg_{0.06}TaS_2$ under $H_{//c}$. $H_{\perp c}$ and $H_{//c}$ indicate the magnetic fields perpendicular and parallel to the preferred orientated *c*-axis, respectively.

7. The effect of structural change on electronic state

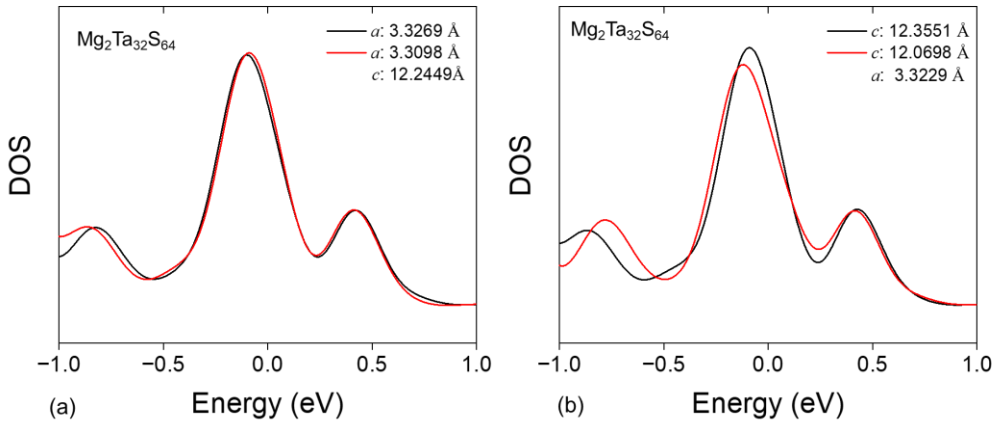


Figure S8. The density of state of $Mg_2Ta_{32}S_{64}$ with described lattice parameters. (a) The black and red lines correspond to the calculated results when the lattice constant *a* is set to 3.3098 Å and 3.3269 Å with fixing the lattice constant *c* at 12.2449 Å. (b) The black and red lines correspond to the calculated results when the lattice constant *c* is set to 12.3551 Å and 12.0698 Å with fixing the lattice constant *a* at 3.3229 Å.

8. Activation energies of migration for monovalent cations

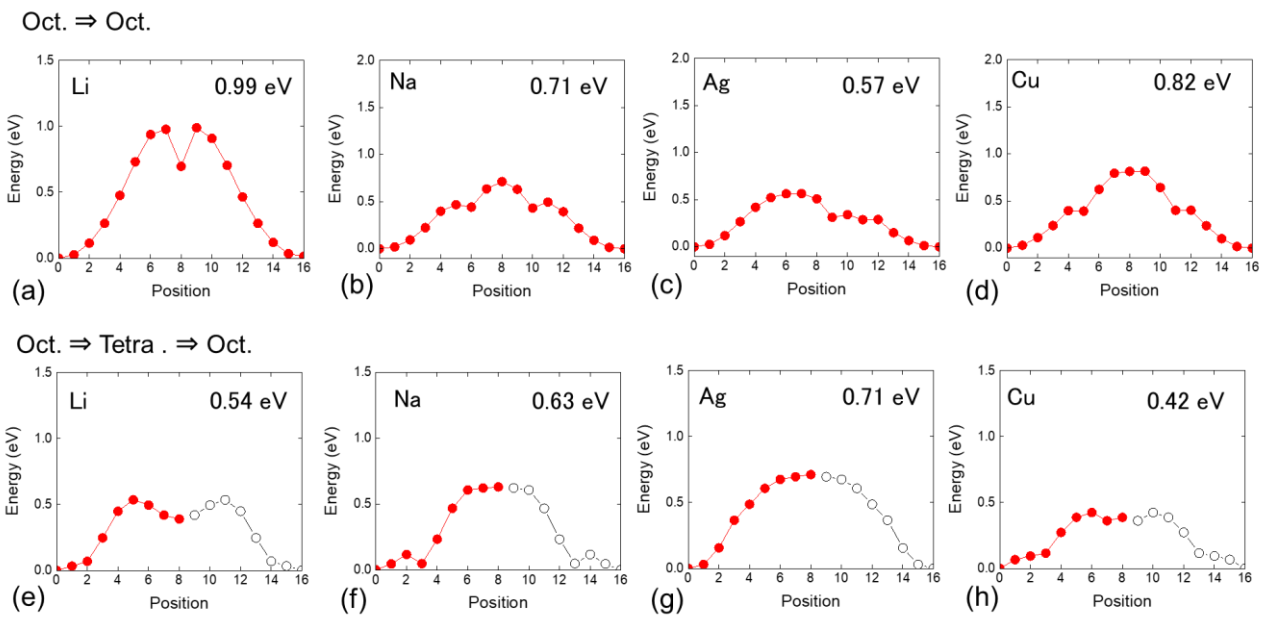


Figure S9. Activation energy of the migration path between octahedral sites for (a) Li, (b) Na, (c) Ag, and (d) Cu and the activation energy of the migration path between octahedral sites via tetrahedral sites for (e) Li, (f) Na, (g) Ag, and (h) Cu.

9. Direct reaction of Mg and TaS_2 and effect of high-pressure synthesis

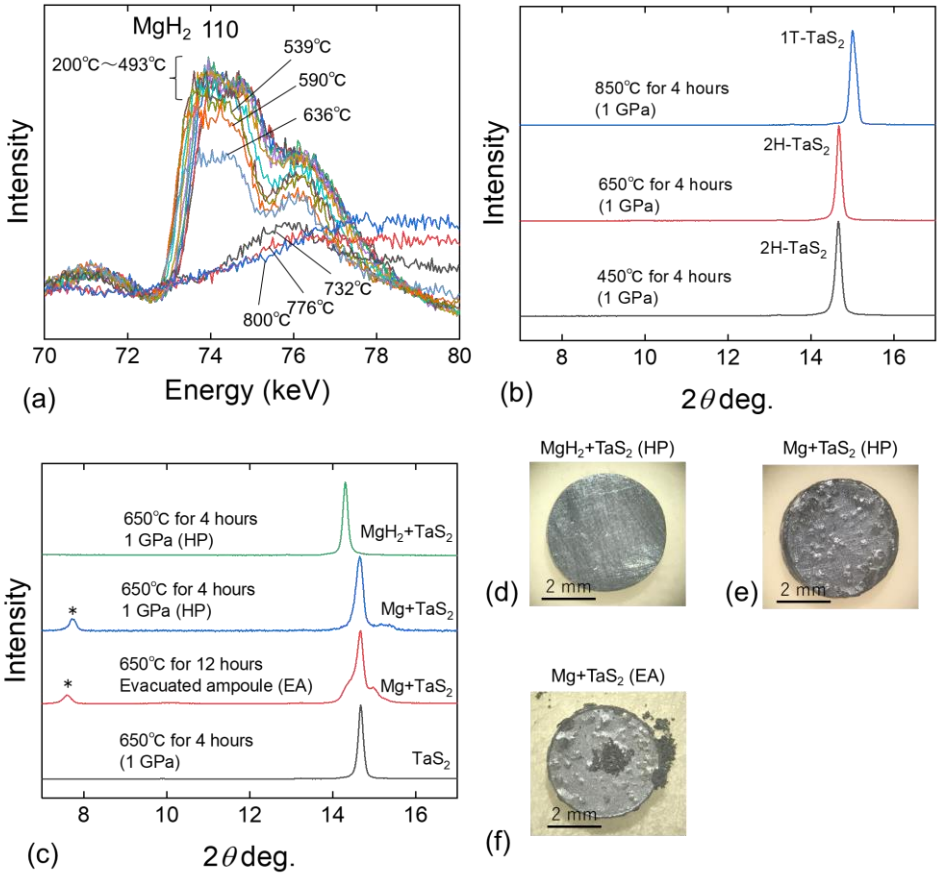


Figure S10. (a) In-situ observation of MgH_2 with increasing temperature at 1GPa using synchrotron energy-dispersive XRD measurements. (b) The XRD peak indexed to 002 for TaS_2 after each treatment at 450, 650, and 850°C under 1 GPa. (c) The XRD peak indexed to 002 for samples after the described treatment condition. Optical images of samples after (d) high-pressure synthesis using MgH_2, (e) high-pressure synthesis using Mg, and (f) sintering in evacuated ampoules using Mg.

Figure S10(a) shows that the peak indexed to 110 gradually attenuated from 539°C and almost disappeared at 732°C. In addition, the high-pressure synthesis at 850°C under 1GPa changes the 2H-TaS_2 to 1T-TaS_2. Based on these results, a heat treatment temperature of 650°C was adopted in this research. Figure S10(c) shows that the direct reaction of Mg and TaS_2 did not show obvious peak shifts and different phases were noted. In addition, as shown in Figures S10(e) and S10(f), when these samples are exposed to air, they gradually decompose from the surface within one hour. The cause of this instability is unknown, but intercalation compounds sometimes incorporate moisture from the air into their structure, resulting in a significant increase in the lattice constant. They may explain the appearance of the peak at a much lower 2θ: 7.7 deg. Thus, significant differences exist between the cases of Mg and MgH_2, and using MgH_2 demonstrated a high effect in promoting Mg intercalation without forming an unstable state.